

BUILD YOUR FIRST

Buddy

A Level 1 build guide

Train your own detector. Make it understand what it sees. Make it talk. Make it do something.

ABOUT 4–5 HOURS

NO ML EXPERIENCE NEEDED



SEES • UNDERSTANDS • TALKS • ACTS

BEFORE YOU START

What are you actually building?

Buddy is a webcam assistant trained on your own stuff. The detector sees. Your code decides what it means.

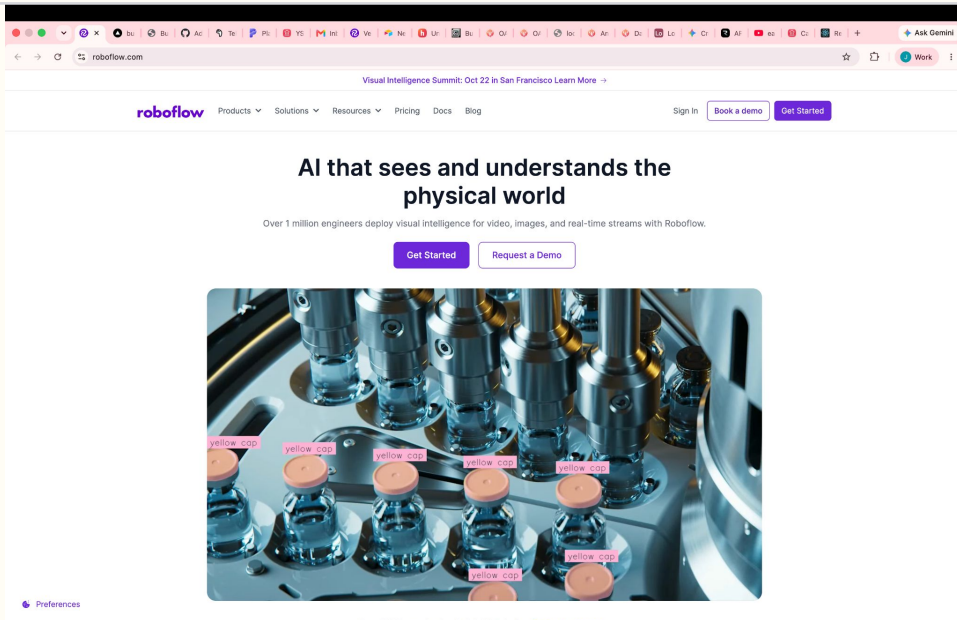


Example: a hand gesture is detected → Buddy says “Volume is going up now” → the page actually raises the volume.

PART 1

Make a Roboflow account

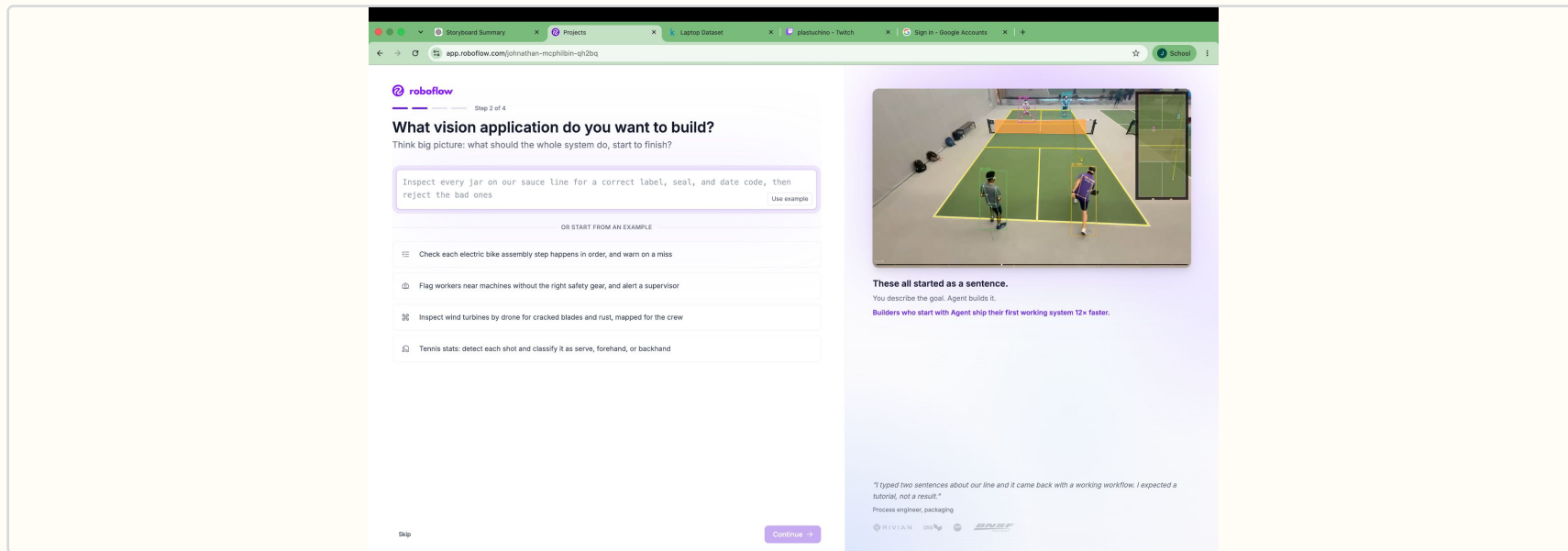
Go to Roboflow and sign in or create an account.



PART 2

Skip the Roboflow Agent

Roboflow may ask four setup questions. You do not need this for Buddy.

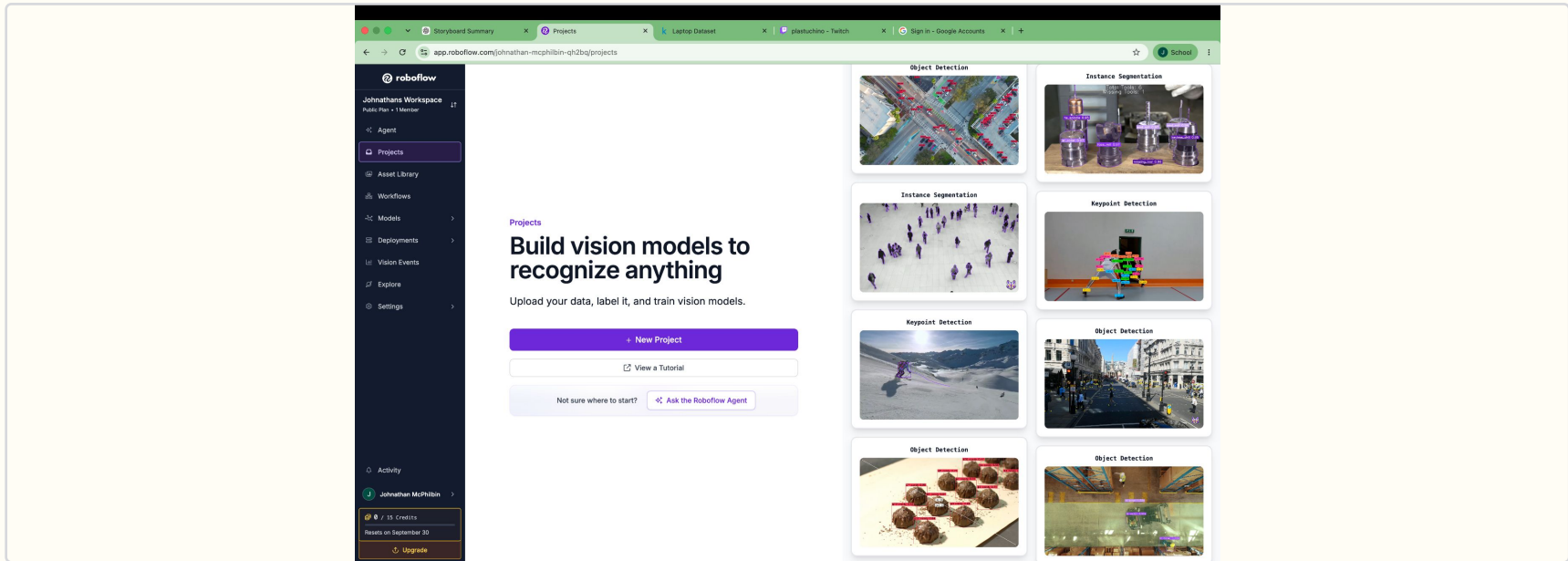


Skip all of the Agent onboarding screens until you land on Projects.

PART 3

Go straight to Projects

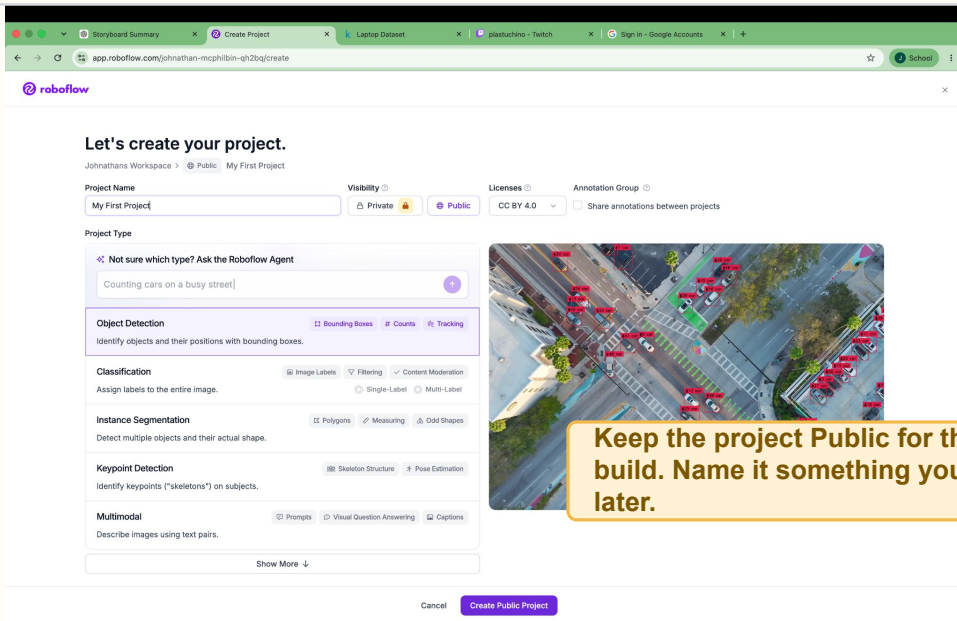
Once you reach the project dashboard, start a normal project.



PART 4

Create an Object Detection project

Buddy needs bounding boxes so it can find multiple things at the same time.



Let's create your project.

Johnathans Workspace > Public > My First Project

Project Name: My First Project

Visibility: ☒ Public ☐ Private

Licenses: CC BY 4.0

Annotation Group: ☐ Share annotations between projects

Project Type

Not sure which type? Ask the Roboflow Agent

Counting cars on a busy street

Object Detection ☒ Bounding Boxes ☐ Counts ☐ Tracking

Identify objects and their positions with bounding boxes.

Classification ☒ Image Labels ☐ Filtering ☒ Content Moderation

Assign labels to the entire image.

☐ Single-Label ☐ Multi-Label

Instance Segmentation ☐ Polygons ☒ Measuring ☐ Odd Shapes

Detect multiple objects and their actual shape.

Keypoint Detection ☐ Skeleton Structure ☐ Pose Estimation

Identify keypoints ("skeletons") on subjects.

Multimodal ☐ Prompts ☐ Visual Question Answering ☐ Captions

Describe images using text pairs.

Show More

Cancel **Create Public Project**

Keep the project Public for the basic Buddy build. Name it something you will recognize later.

PART 5

Pick 3 to 5 things Buddy should know

Start small. You can always make a more ridiculous Buddy later.

A light green rounded rectangle with a dark green border.

KEYS

A light purple rounded rectangle with a dark purple border.

PHONE

A light green rounded rectangle with a dark green border.

WALLET

A light purple rounded rectangle with a dark purple border.

BAG

Take roughly 30 to 50 photos per object.

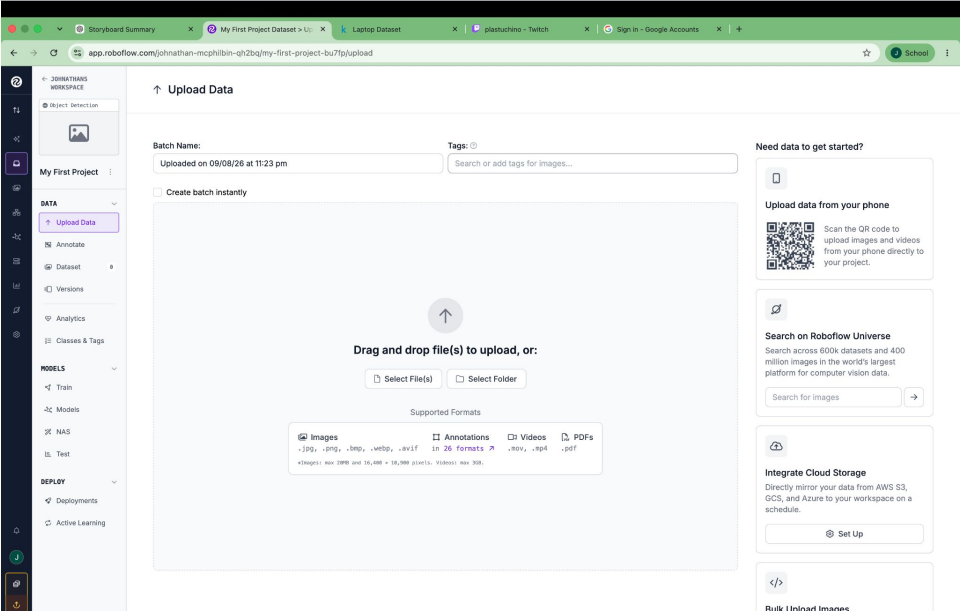
- ✓ Different angles and distances
- ✓ Different lighting and backgrounds
- ✓ Put multiple objects in the same photo
- ✓ Use photos you took yourself

Do not take 50 nearly identical photos on the same desk. Variety is what makes the detector useful.

PART 6

Upload your photos

Open Upload Data and add the photos you just took.

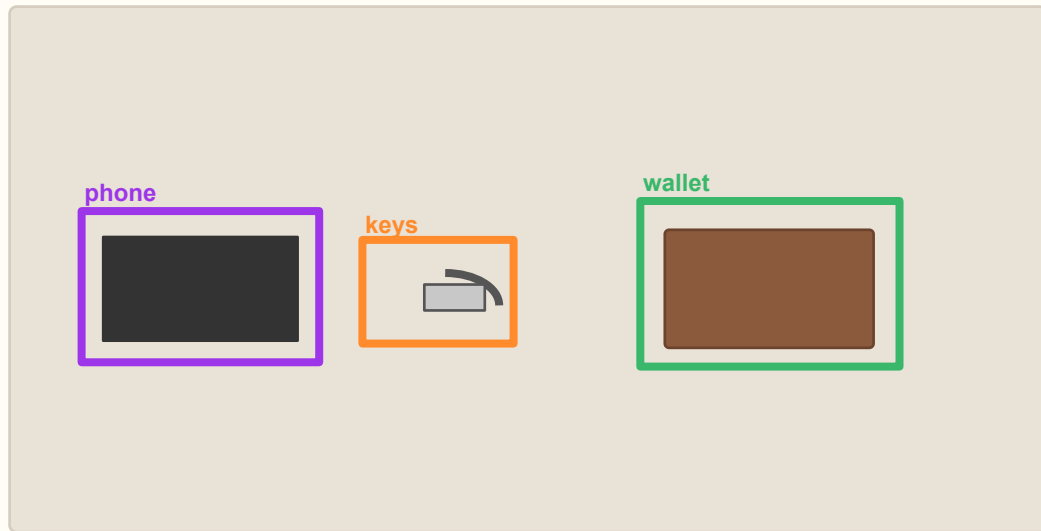


The screenshot shows the Roboflow web interface for uploading data. The browser address bar displays `app.roboflow.com/johnathan-mcpiblin-qh2bq/my-first-project-bu7fp/upload`. The left sidebar contains navigation options: MONITORSPACE, My First Project, DATA (with 'Upload Data' highlighted), Annotate, Dataset, Versions, Analytics, Classes & Tags, MODELS (with Train, Models, NAS, and Test), and DEPLOY (with Deployments and Active Learning). The main content area is titled 'Upload Data' and includes a 'Batch Name' field with the value 'Uploaded on 09/08/26 at 11:23 pm', a 'Tags' field, and a 'Create batch instantly' checkbox. A large dashed box in the center contains an upward arrow and the text 'Drag and drop file(s) to upload, or:'. Below this, there are buttons for 'Select File(s)' and 'Select Folder', and a 'Supported Formats' section listing Images, Annotations, Videos, and PDFs with their respective file extensions and dimensions. On the right side, there are three sections: 'Need data to get started?' with a QR code for mobile upload, 'Search on Roboflow Universe' with a search bar, and 'Integrate Cloud Storage' with a 'Set Up' button. At the bottom right, there is a 'Bulk Upload Images' section.

PART 7

Annotate every object

Draw a tight box around every thing Buddy should recognize and give it the right label.



If the photo contains 3 things, it needs 3 boxes.

- ✓ Box every relevant object
- ✓ Keep boxes reasonably tight
- ✓ Use the same class name every time

phone ≠ Phone ≠ iphone. Pick one name and stick to it.

PART 8

Create a dataset version and train

Once everything is labelled, turn the dataset into a model. Roboflow handles the neural network bit.

1

Open Dataset / Versions

Create a new version from your annotations.

2

Generate the version

Default settings are fine for your first Buddy.

Train a model

Start training from the version you just made.

4

Wait for training to finish

Then test it on images it has not seen before.

Do not use a stock dataset for the base Buddy. The detector should be trained on photos you collected and labelled yourself.

PART 9

Test the detector before writing Buddy logic

Your model should be able to recognize multiple things at the same time.

CHECKPOINT

Can Buddy correctly find at least two of your classes in the same image?

YES → KEEP GOING

NO → ADD MORE VARIED PHOTOS

If it keeps confusing objects, the fix is usually better data, not more code.

YOU WILL NEED

Project slug
Model version
Publishable key

Copy the browser deployment details
Roboflow gives you for your trained model.

Never put a private Roboflow API key in frontend code. Use the publishable key intended for browser inference.

PART 10

Connect your model to the Buddy starter

The starter handles webcam access, model loading, and drawing boxes so you can focus on the actual Buddy behavior.

```
const PROJECT_ID = "your-project-slug";  
const MODEL_VERSION = 1;  
const PUBLISHABLE_KEY =  
  "your-publishable-key";
```

Replace the three placeholders with the details from your Roboflow model.

<code>detection.ts</code>	runs the model
<code>buddy.ts</code>	decides what detections mean
<code>speech.ts</code>	makes Buddy talk
<code>+page.svelte</code>	shows the camera and UI

When you run the starter and turn on your webcam, you should see live bounding boxes before you move on.

PART 11

Make Buddy understand what it sees

The model tells you what is visible. Your code decides what that situation means.

```
const seesKeys = detectedClasses.includes("keys");  
const seesPhone = detectedClasses.includes("phone");  
const seesWallet = detectedClasses.includes("wallet");  
const seesBag = detectedClasses.includes("bag");
```

Then turn the state into rules:

```
if (seesKeys && seesPhone && seesWallet && seesBag) {  
  response = "You are good to go.";  
} else if (seesKeys && seesPhone && !seesWallet) {  
  response = "You forgot your wallet.";  
}
```

THE IMPORTANT BIT

Detection is not the project.

What you make the detections mean is the project.

PART 12

Make Buddy talk

Use the browser's built-in speech synthesis. No external voice service is required for Level 1.

```
function say(text) {  
  const u = new SpeechSynthesisUtterance(text);  
  speechSynthesis.cancel();  
  speechSynthesis.speak(u);  
}
```

Do not call this on every webcam frame or Buddy will never shut up.

- ✓ Wait until the detected combination is stable for about 1 second
- ✓ Speak once
- ✓ Do not repeat until the state changes
- ✓ Cancel old speech before a new response

PART 13

Make Buddy do something

Talking is only half of it. At least one detection must cause a real action in your project.

```
if (seesThumbsUp) {  
  say("Volume is going up now.");  
  increaseVolume();  
}
```

VOLUME

change audio

TIMERstart or stop
something**SCORE**count what Buddy
sees**PAGE**change the
interface**ALERT**

show a warning

HARDWAREtrigger LEDs or
motors

If Buddy only says "I see a phone," you are not done yet. Make the detection change something.

PART 14

Give Buddy at least 6 different outcomes

Use combinations, missing objects, counts, or gestures so the assistant actually has behavior.

ALL 4 ITEMS

“You are good to go.”

NO WALLET

“You forgot your wallet.”

KEYS ONLY

“That is not enough.”

NOTHING

“You prepared absolutely nothing.”

THUMBS UP

“Volume is going up now.”

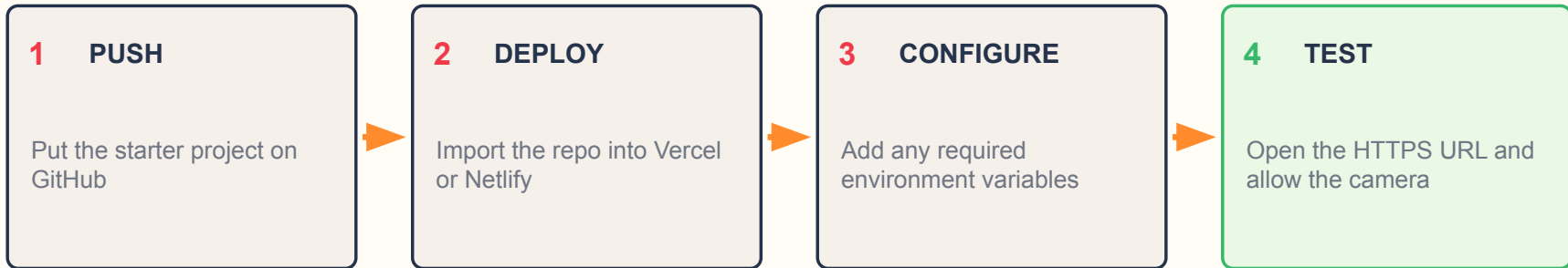
3RD SNACK

“That is snack number three.”

The exact six are up to you. They just need to be genuinely different outcomes, not six ways of saying “I see X.”

Put Buddy on the internet

Your submission is a real URL where Buddy works. No video demo is required for this tutorial.



Your final link should look something like:

`https://your-buddy.vercel.app`

Test the deployed URL, not just localhost. Camera access should work over HTTPS.

YOU ARE DONE

Level 1 complete



If every box below is true, you have shipped a Basic Buddy.

- ✓ Your own Roboflow dataset
- ✓ Live object or gesture detection
- ✓ Multiple detections or meaningful gesture classes
- ✓ At least 6 different outcomes
- ✓ Buddy speaks responses out loud
- ✓ At least one real action happens
- ✓ A real deployed HTTPS URL
- ✓ Hackatime project connected

IT SEES. IT UNDERSTANDS. IT TALKS. IT DOES STUFF.

Next: spend your Bits on upgrades and push Buddy beyond Level 1.